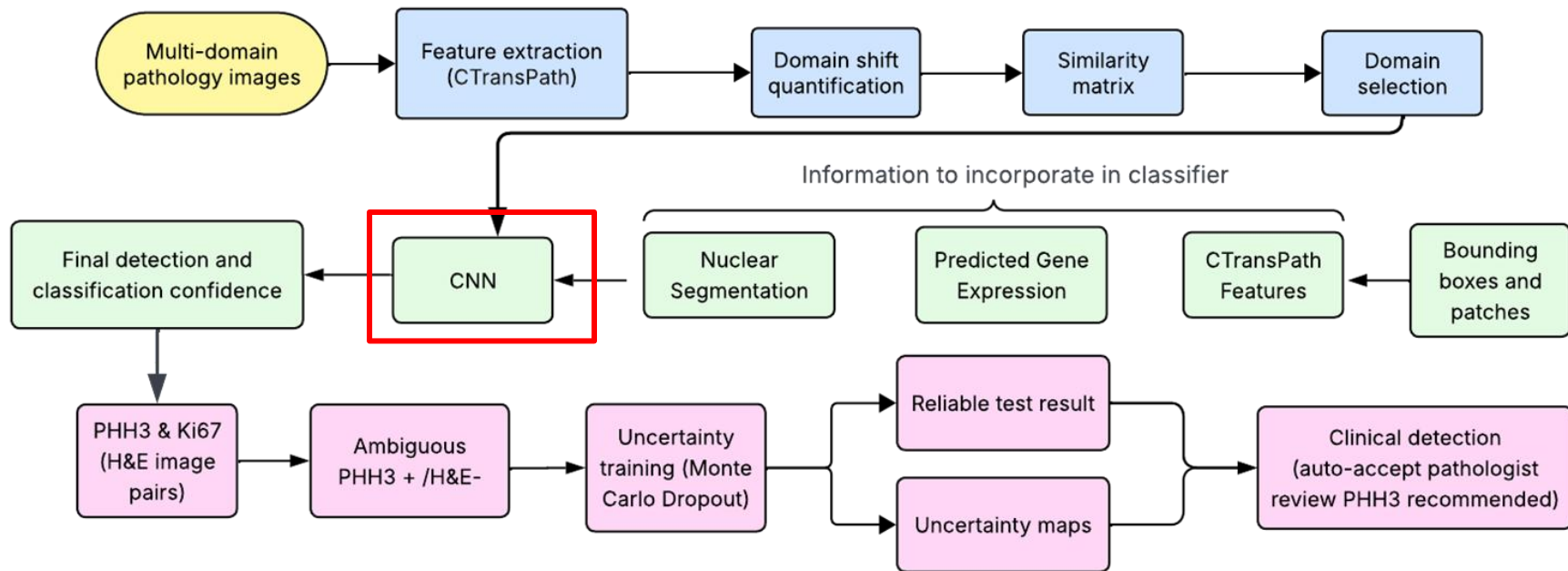




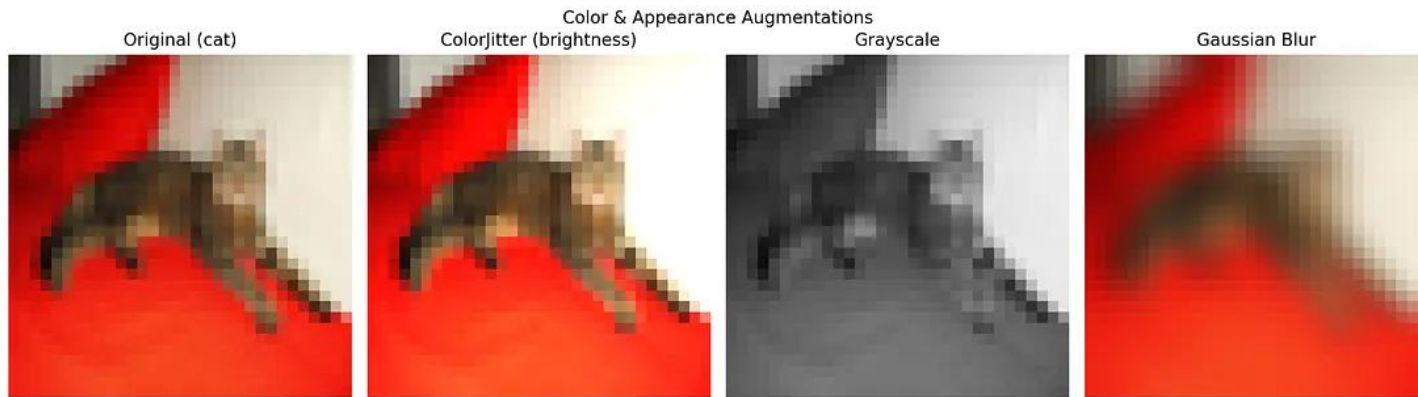
Data Augmentation

Naina Misra | Yang Cao | Lillian Lam | 02/26/2025

Pipeline

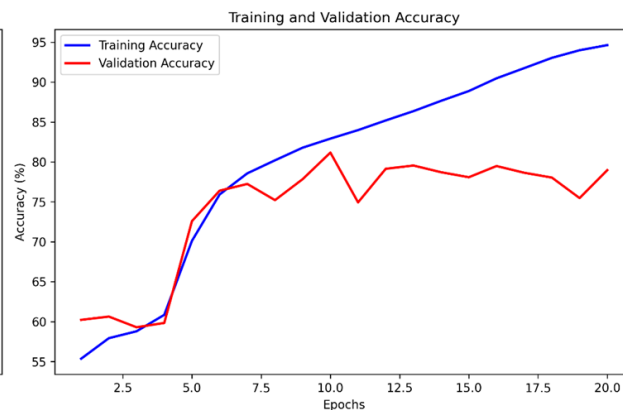
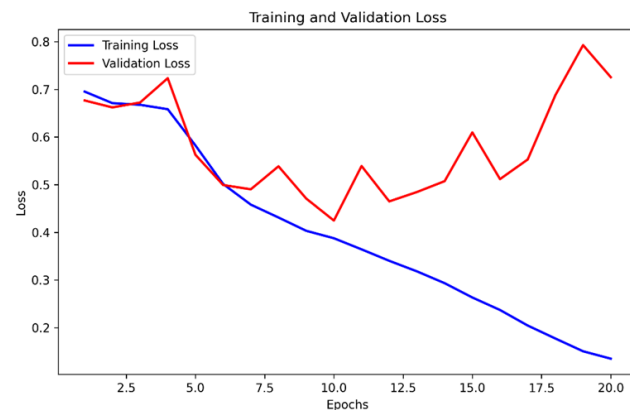


Data Augmentation



A technique to work with limited data (a common problem in machine learning) by expanding the dataset via image modifications. Through introducing greater image variety, model performance can be improved.

Initial Model (No Augmentation)



Confusion Matrix	Predicted non-mitotic	Predicted mitotic
Actual non-mitotic	TN	FP
Actual mitotic	FN	TP

Confusion Matrix:

[[2256 341]

[532 1507]]

Key Metrics:

Sensitivity (Recall): 0.739

Specificity: 0.869

Precision: 0.815

F1-Score: 0.775

Best Test Accuracy: 81.17%

Color Jitter

100% images augmented

Confusion Matrix:

```
[[2221 376]
```

```
 [ 444 1595]]
```

Key Metrics:

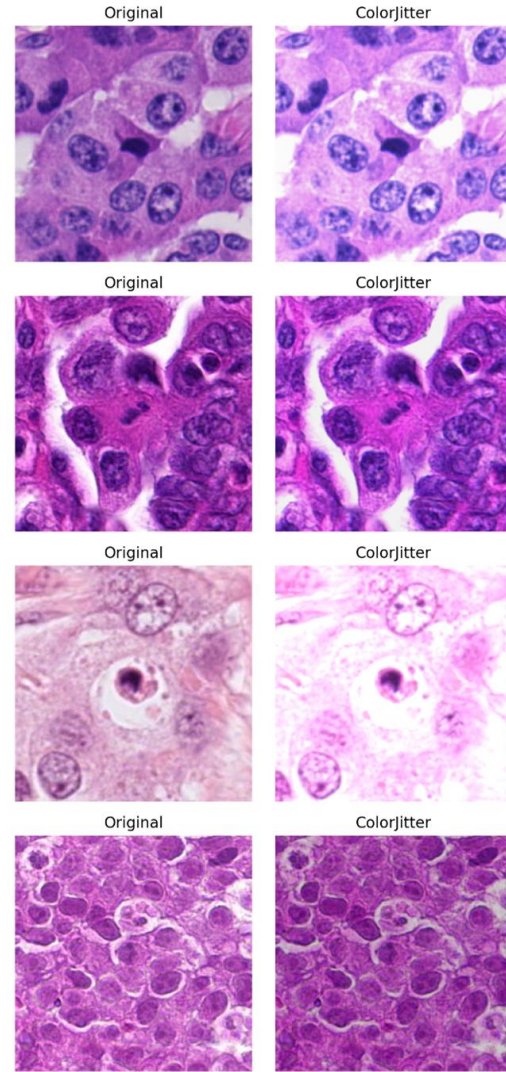
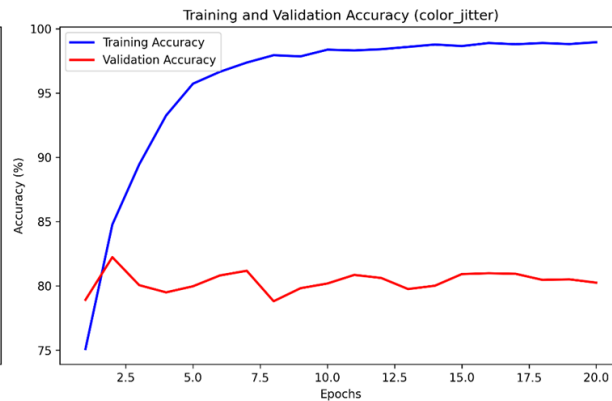
Sensitivity (Recall): 0.782

Specificity: 0.855

Precision: 0.809

F1-Score: 0.796

Best Test Accuracy: 82.23%



Grayscale

30% images augmented

Confusion Matrix:

```
[[2221  376]  
 [ 444 1595]]
```

Key Metrics:

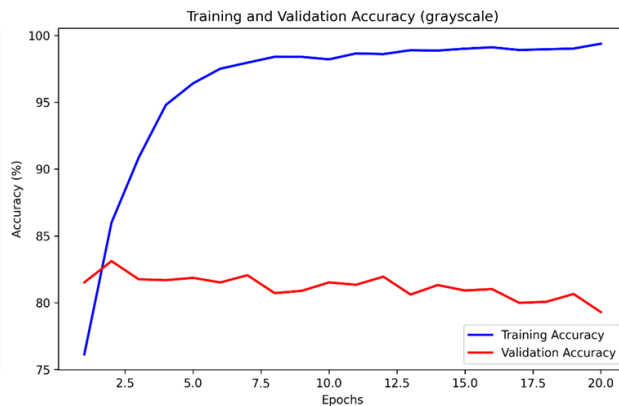
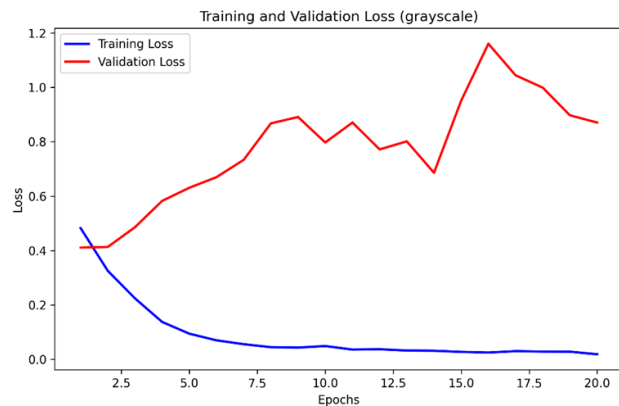
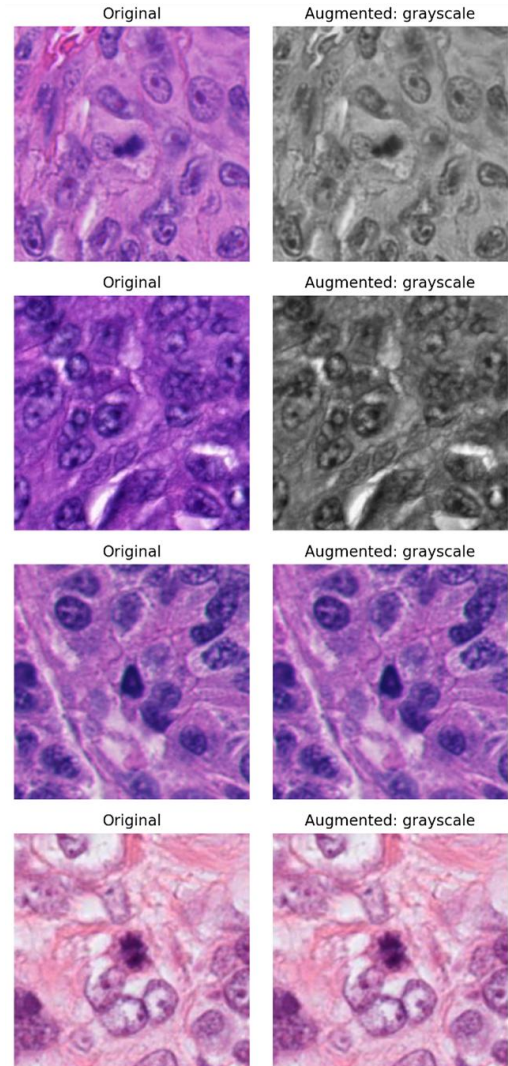
Sensitivity (Recall): 0.782

Specificity: 0.855

Precision: 0.809

F1-Score: 0.796

Best Test Accuracy: 83.11%



Gaussian Blur

30% images augmented

Confusion Matrix:

```
[[2221  376]
```

```
 [ 444 1595]]
```

Key Metrics:

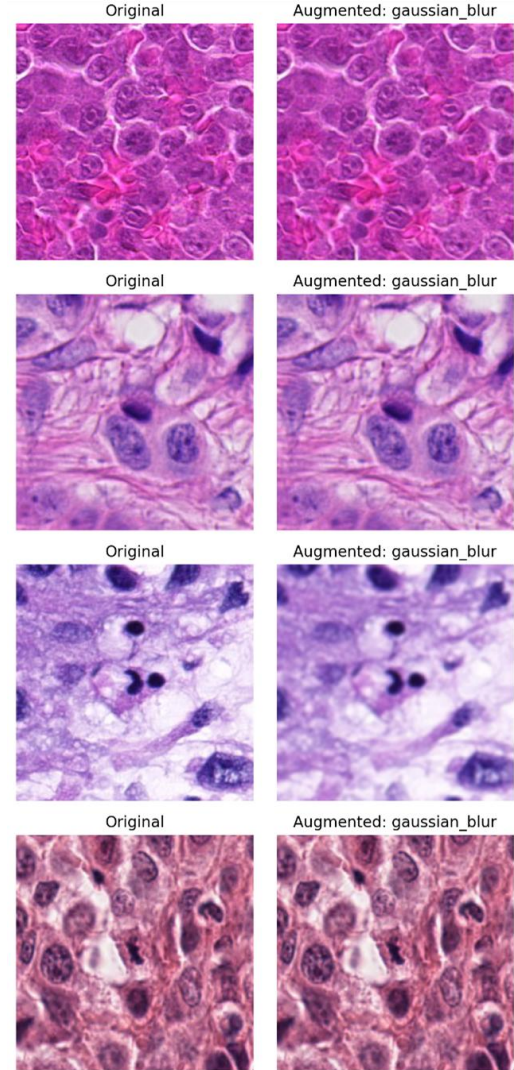
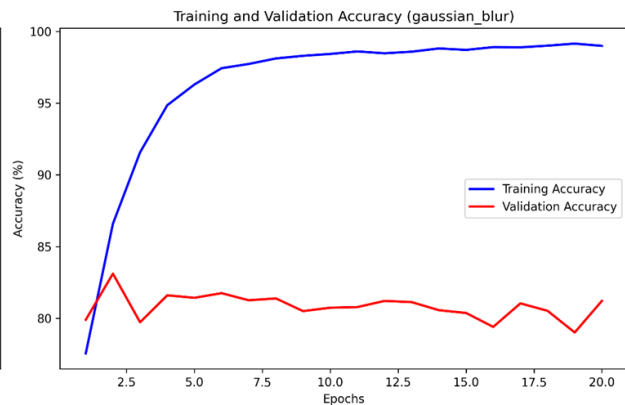
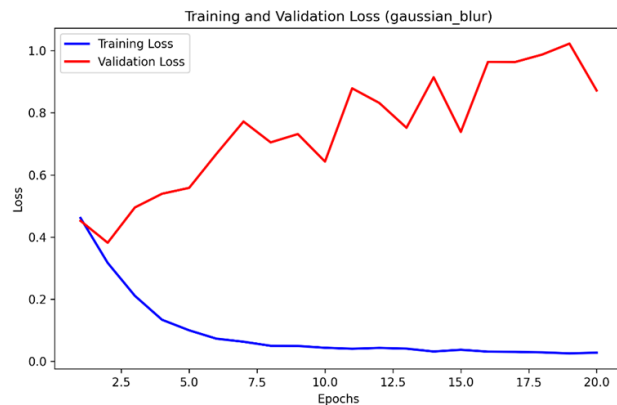
Sensitivity (Recall): 0.782

Specificity: 0.855

Precision: 0.809

F1-Score: 0.796

Best Test Accuracy: 83.11%



Cutmix

50% images augmented

Confusion Matrix:

```
[[2221  376]
 [ 444 1595]]
```

Key Metrics:

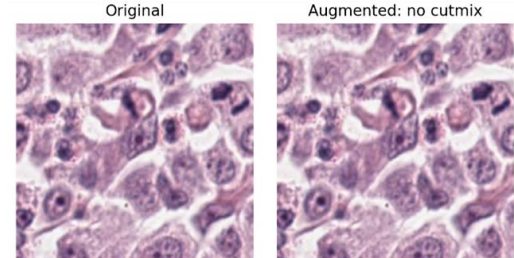
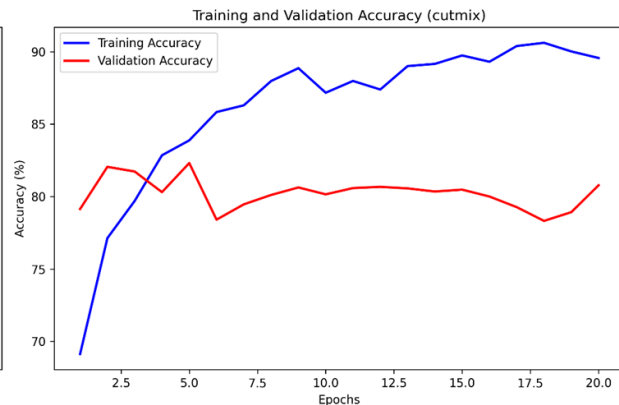
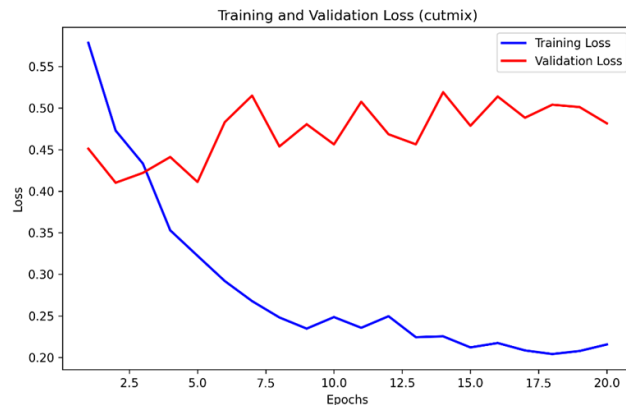
Sensitivity (Recall): 0.782

Specificity: 0.855

Precision: 0.809

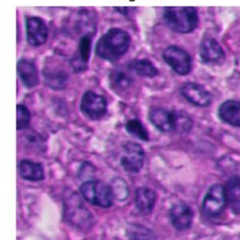
F1-Score: 0.796

Best Test Accuracy: 82.31%

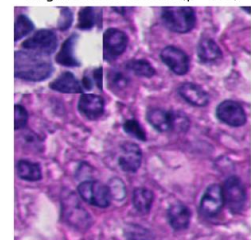


Original

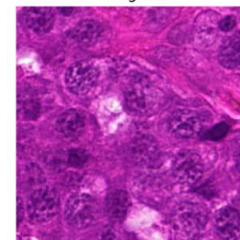
Augmented: no cutmix (p=0.5)



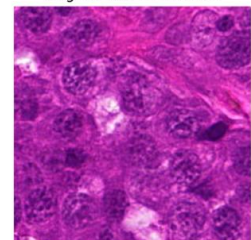
Original



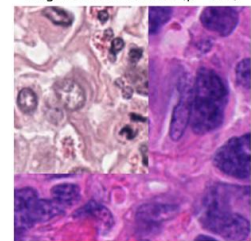
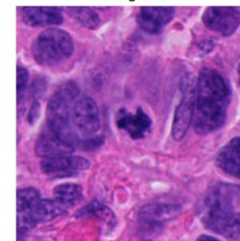
Augmented: no cutmix



Original



Augmented: cutmix (p=0.5)



All

Randomized
augmentation of all
types

Confusion Matrix:

```
[[2265 332]
```

```
[ 529 1510]]
```

Key Metrics:

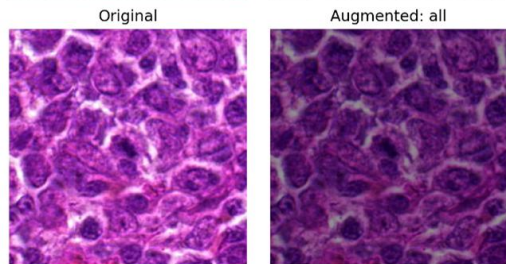
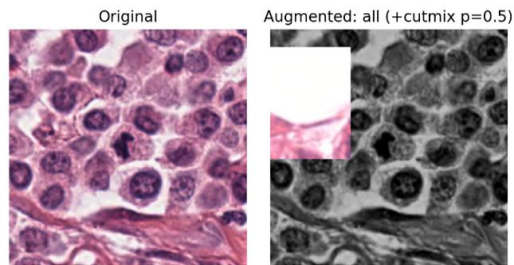
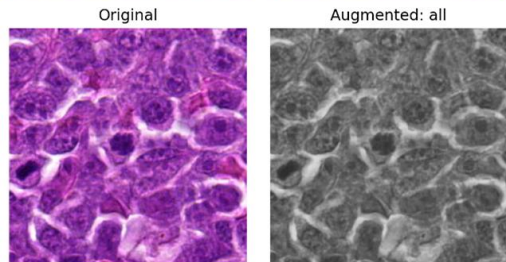
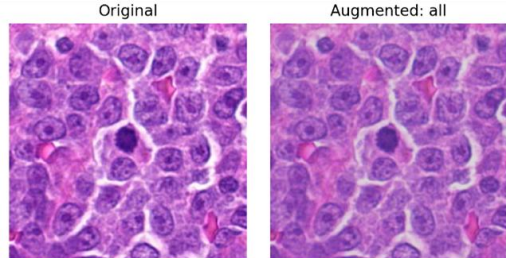
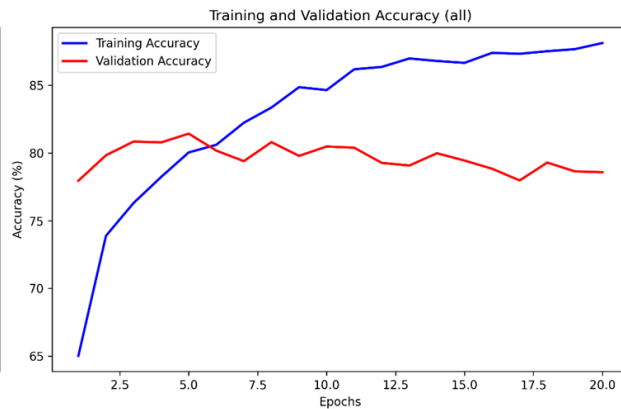
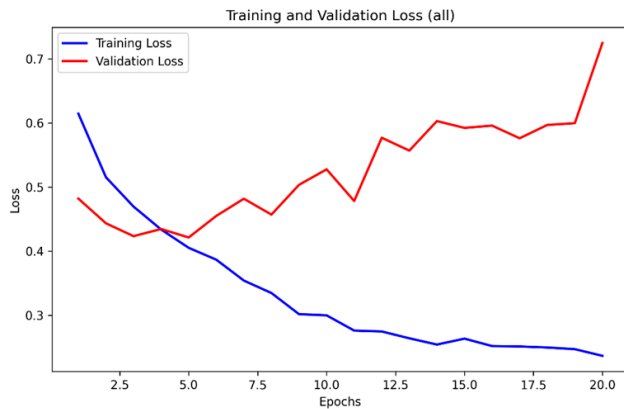
Sensitivity (Recall): 0.741

Specificity: 0.872

Precision: 0.820

F1-Score: 0.778

Best Test Accuracy: 81.43%



Limitations

Fast overfitting (5-6 epochs)

Training accuracy is high, validation plateaus around 80%

Noise at the patch level

~ 400 WSIs → 21k patches

Potentially reaching model limit

Potential Solutions

- Reduce batch size/learning rate
- Tune augmentation parameters
- Test other augmentation types and identify most effective
- Extracting biological features from WSIs

Next Steps

- Continue training with Resnet50 classifier
- Compare accuracy and F1 score to the ranked WSI images
- Train on only Hematoxylin part of the WSI
- Try a different H&E to Spatial Transcriptomics packages

Thank you!